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AN ANALYSIS OF THE STRESSES INCURRED IN PIPE DURING LAYING OPERATIONS

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ABSTRACT

It is common practice during the construction of underground pipelines that pipe sections are welded together at surface level and then lowered into a pre-dug trench. Lowering of the pipeline into the trench incurs bending and/or torsional stresses that are not typically considered during the design of the pipeline. Current industry guidelines recommend the pipelines be lowered into the trenches with care in order to avoid cracking of the pipe, or yielding that could lead to residual stresses and distortion. There are currently no formal guidelines on how to minimize these stresses, or estimates of the magnitude of stresses associated with these operations.

The stresses incurred during the laying of pipe can be estimated using simple beam bending and torsion equations. One can simplify the analysis by assuming the pipeline is either a cantilever with a given load or deflection at one end, or, a beam with given loads, deflections and rotations at both ends. The choice of boundary conditions influences the calculated stresses, and can be considered appropriate for different pipe laying operations. If a pipeline is moved laterally during the laying operations one must consider the associated torsional stresses. The combination of bending and torsional stresses will lead to different stress states on different sides of the pipe, some of which may result in cracking or yielding.

The goal of this work is to develop a framework for estimating the stresses incurred in pipe during laying operations, as a function of pipe length, deflection, diameter and wall thickness. This will allow guidelines to be developed that will allow construction crews to ensure that pipelines are laid with minimal damage. A variety of analyses has been considered during the course of this work and is presented in this paper. It is expected that discussion of these analyses with industry representatives will lead to improved construction protocols.

INTRODUCTION

It is common practice during the construction of buried onshore pipelines to weld several joints of pipe together before lowering them into a pre-dug trench. The construction process consists of several stages. Crews begin the process by clearing the right-of-way for the pipeline. The trench is dug using a trencher or back-hoe. Joints are then welded together, using a variety of techniques. The welds can then be coated for corrosion resistance. Depending on the environment, rock protection or buoyancy controls may be necessary. The pipe can then be lowered into the pre-dug trench using a series of sidebooms. Crews finally cover the newly laid pipeline with backfill.

The stresses in a pipeline experienced during laying operations may be significant. The pipeline is subjected to both bending stresses and torsional stresses. The bending stresses are produced by a combination of the (upward) sideboom lifting forces and the (downward) weight of the pipe. Both the height the pipeline is lifted and the spacing between sideboom lifting points will affect the magnitude of the bending stress. In addition, there may be torsional stress in the pipe as it is moved laterally from the side of the trench to the bottom center of the trench. Improper lifting and lowering protocols may lead to stresses that damage the pipeline.

A review of various industry standards [1-5] indicated that there is little in the way of formal guidance available for pipe lowering. ASME B31.4 and B31.8 state that “it is important that stresses imposed on the pipeline by construction be minimized”, and advise that “care be exercised so as not to impose undue stress in the pipe”. CSA Z662 states that “piping shall be lowered into the ditch in a manner that avoids excessive stress levels.”

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There is guidance available for calculating the stress experienced during movement of in-service pipelines [6]. API 1117 provides guidelines for the calculation of the axial stresses resulting from in-service pressure and temperature changes, and the axial and bending stresses resulting from movement of pipelines to accommodate obstructions. This document provides various equations that can be used as a framework for calculation of the stresses involved in pipe laying operations, though there are different features of the bending that need to be addressed.

There is guidance available for calculating the stresses resulting during the laying of off-shore pipelines [7]. Lowering off-shore pipelines onto the ocean floor tends to be a more complex operation than that of on-shore pipelines. Off-shore pipeline lowering is influenced by the depth of water, seabed topography, weather and wave action during the laying operation, and the difficulties in welding and storage of pipe on board the laying barge. These are not issues faced by the on-shore pipeline construction crews. However, the associated guidelines and standards also provide guidance that can be used for the current work.

The objective of this work is to estimate the stresses experienced by pipelines during onshore laying operations. This may then lead to the development of improved construction protocols, and reduction of damage occurred to pipelines during construction.

The stresses experienced in the pipeline during lowering operations can be estimated using classical elastic beam bending and torsional analyses. The various stages of the laying operation lead to different curvature in the pipeline that can be simulated using beams with various loadings and/or boundary conditions. The majority of this work is based on beam bending calculations. A lesser component of the work is based on torsional analysis. Various scenarios are considered. In each case the applicability of the analysis is discussed with reference to a particular pipe geometry, curvature and/or rotation angle.

In each case discussed, the stresses are estimated as a function of the pipe geometry (diameter and wall thickness), and the lift height, and span or rotated length. Construction crews can therefore easily determine the maximum height a pipe can be lifted or the maximum span a pipe can remain unsupported, without any significant risk of pipe yielding. Note that the stresses calculated are nominal values and the safety factors associated with stress concentrators should be considered as applicable. In addition, various assumptions have been made to simplify the analyses, so the estimates provided should not be considered rigorous solutions. These assumptions and simplifications are discussed as applicable.

ANALYSIS

The analysis considers six different pipe geometry and loading cases:

Case 1. Beam with end lift point.
Case 2. Beam with central lift point.
Case 3. Beam sagging under its' own weight.
Case 4. Fixed span.
Case 5. Freely supported span.
Case 6. Torsion of the pipe.

The relevance and applicability of these six geometries is discussed in turn the sections below. In each case a schematic and a plot of the analysis results are provided. The plot can be used to directly estimate the nominal stresses associated with a given laying operation.

The final exercise of the work is to provide a very simple example that can demonstrate how the work can be applied to a field situation. Each one of the above mentioned cases approximates the loading and deflection of a pipeline during laying operations.

CASE 1. Beam with end lift point.

This case is applicable when a sideboom lifts one end of the pipeline off the ground and the other end is resting on the ground due to its own weight. It would be applicable either at the start of the construction when the sideboom initially lifts the pipeline off the ground (or support structures), or at the end of the construction when the side boom lays the final length down in the pre-dug trench. A schematic is given as Figure 1.

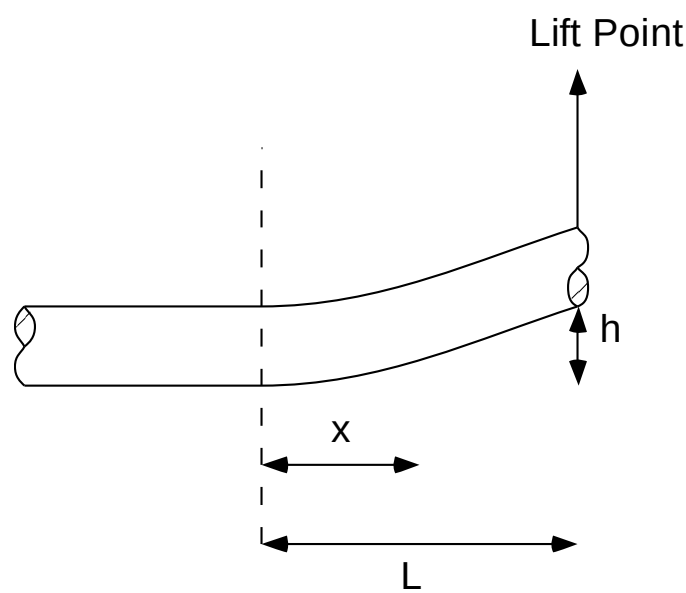

Figure 1 Schematic of Case 1, beam with end lift point.

The case is treated by cantilever beam analysis. There are two loads to be considered; the point load provided by the sideboom that lifts one end of the pipe, and the distributed weight of the pipe itself. In order to simplify the calculations, it has been assumed that the lifting force is equal to the total

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weight of pipe lifted off the ground. This is not rigorously correct, as some of this weight will be at the ground end of the pipe. However, the proportion and distribution of this weight will vary with excess length of pipe lying on the ground. The above assumption provides a conservative estimate of the stresses.

Using simple mechanics (or by combining standardized solutions [8]) the moment acting on the pipe can be calculated as:

$$M(x) = \frac{1}{2} wL^2 - \frac{1}{2} wx^2 \qquad (1)$$

Where M is the moment acting on the pipe, w the distributed weight per unit length of the pipe, L the length of pipe lifted off the ground, and x the distance along the pipe from the point where the pipe last touches the ground. The weight per unit length of the pipe varies with diameter and wall thickness:

$$w = \frac{\pi}{4}\left(OD^2 - ID^2\right) \cdot \rho \qquad (2)$$

Where OD is the outer pipe diameter, ID the inner pipe diameter and ρ the material density. For the purpose of these analyses, it has been assumed that all pipes under consideration are steel pipes with a density of 0.284 lb/in^3.

The bending stress in the pipe is given by:

$$\sigma = \frac{My}{I} \qquad (3)$$

Where σ is the bending stress, y the distance from the neutral axis and I the second moment of area. The distance from the neutral axis is given by (OD/2), as the outer fiber stress is the maximum stress and therefore the stress of interest. The second moment varies with pipe diameter and wall thickness:

$$I = \frac{\pi}{64}\left(OD^4 - ID^4\right) \qquad (4)$$

The deflection of the pipe can be calculated by integrating the moment twice and dividing by the modulus and second moment of area:

$$\delta(x) = \frac{1}{EI}\left[\frac{1}{4} wL^2 x^2 - \frac{1}{24} wx^4\right] \qquad (5)$$

Where δ is the deflection and E the modulus. This calculation assumes the pipe is horizontal at the point where it last touches the ground.

The length of pipe that lifts off the ground is a function of the lift height. Evaluating the deflection at the lift point and re-arranging yields:

$$L = \left[h \cdot \frac{24}{5} \cdot \frac{EI}{w}\right]^{1/4} \qquad (6)$$

Where h is the lift height (deflection) of the end of the pipe, i.e. at the lift point. The equation demonstrates that the length of pipe lifted off the ground is a function of the lift height. This is intuitively correct.

It is of interest in this study to calculate the maximum stress in the pipe as a function of lift height off the ground. To achieve this, the stress must be evaluated at the point the pipe lifts off the ground $x = 0$, but the deflection must be evaluated at the lift point $x = L$. Evaluating and combining the above equations yields:

$$\sigma_{\max} = \frac{1}{2} w \frac{y}{I}\left[h \cdot \frac{24}{5} \cdot \frac{EI}{w}\right]^{1/2} \qquad (7)$$

Equation (7) demonstrates that the higher the pipe is lifted off the ground, the greater the maximum bending stress. This is intuitively correct. The above equation is the development of a very simple tool to estimate the bending stress resulting from the lifting of one end of a length of pipe. The relationship is plotted as Figure 2 for three common pipe geometries.

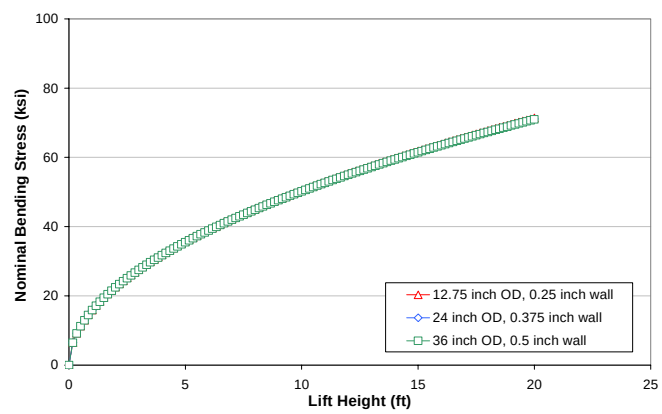

Figure 2 Plot of Case 1, beam with end lift point, calculated nominal bending stresses.

The figure demonstrates that the maximum bending stress is not sensitive to pipe geometry for this case. All three of the curves overlap. Note that the geometric factors (OD and ID) influence the distributed weight, second moment, and distance from neutral axis. The diameter and wall thickness combinations selected appear to have a "self-balancing" effect on each other. The larger pipe is heavier, but stress is

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distributed over the larger cross-section. The smaller pipe is lighter, but the stress is confined to a smaller cross-section.

The stress estimated above can be compared to the yield strength of the pipe steel of interest. A "critical" lift height can be determined that should not be exceeded. Given that the stresses calculated are nominal values, an appropriate safety factor should be considered.

CASE 2. Beam with central lift point.

This case is applicable when the sideboom lifts the pipe from a point near the center of the length, such that both ends are lying on the ground. Bending stresses are both at the sideboom lift point, and at the two points where the pipe touches down on the ground. A schematic is given as Figure 3.

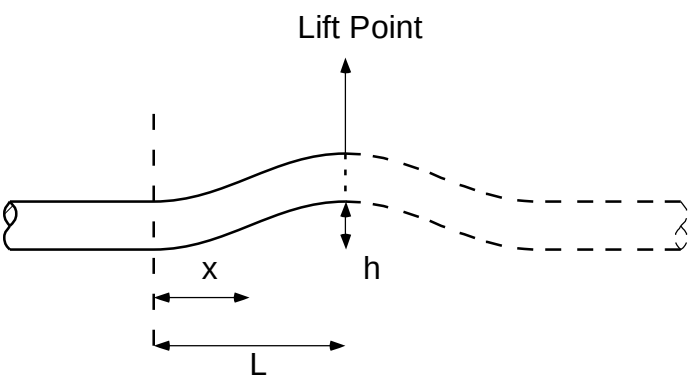

Figure 3 Schematic of Case 2, beam with central lift point.

Like Case 1, this case is also treated as a cantilever, but for simplicity. Advantage is taken of symmetry and only half of the beam is considered. The analysis provided above is followed, but boundary conditions are prescribed such that the end of the cantilever beam is horizontal at both ends. This is enforced as a calculated constant during the integration of the moment equation. Effectively, only half of the beam is analyzed, but the symmetry provides an appropriate solution. The analysis yields:

$$\sigma_{\max} = \frac{1}{4} w \frac{y}{I} \left[h \cdot 24 \cdot \frac{EI}{w} \right]^{1/2} \qquad (8)$$

Note that the equation differs only in the constants involved. The relationship is plotted as Figure 4 for three common pipe geometries. As with Figure 2, the three curves ovelap. A "critical" lift height can be determined that should not be exceeded.

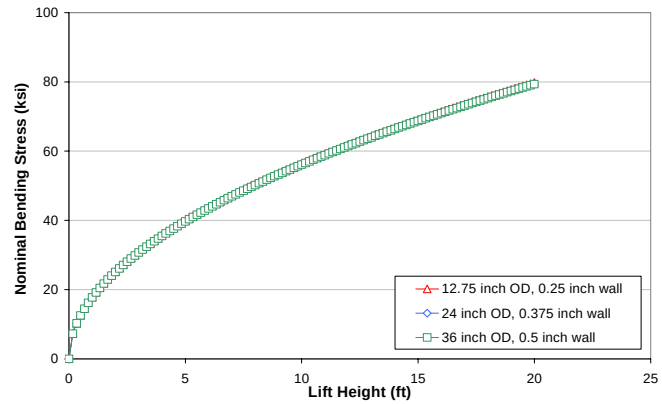

Figure 4 Plot of Case 2, beam with central lift point, calculated nominal bending stresses.

CASE 3. Cantilever sagging under own weight.

This case is applicable when the sideboom lifts the pipe , but with the loading point away from the end of the pipe. The excess length of pipe sags under its own weight, leading to a bending stress at the lift point. This case could be applicable at various stages of the construction process. A schematic is given as Figure 5.

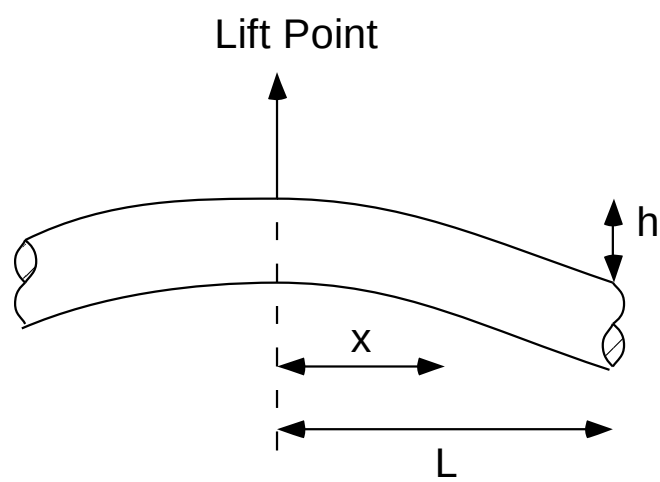

Figure 5 Schematic of Case 3, cantilever sagging under own weight.

The case is also treated by cantilever beam analysis. There is only one load to be considered; the distributed weight of the pipe itself. Note that for this case the length of excess pipe that can sag under is own weight is limited by the lift height. If the lift height is too low or the length of excess pipe is too long, then the end of the pipe would lie on the ground, and Case 2 would be more applicable.

In this case, a slightly different approach is taken, and the critical dimension is the length. The bending stress is calculated

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directly as a function of pipeline length. There is no determination of a critical lift height. The maximum bending stress is given by simple cantilever analysis:

$$\sigma_{\max} = \frac{1}{2} w L^2 \frac{y}{I} \qquad (9)$$

This calculation is similar to Cases 1 and 2, except that the equation is in terms of pipe length. The relationship is plotted as Figure 6 for three common pipe geometries.

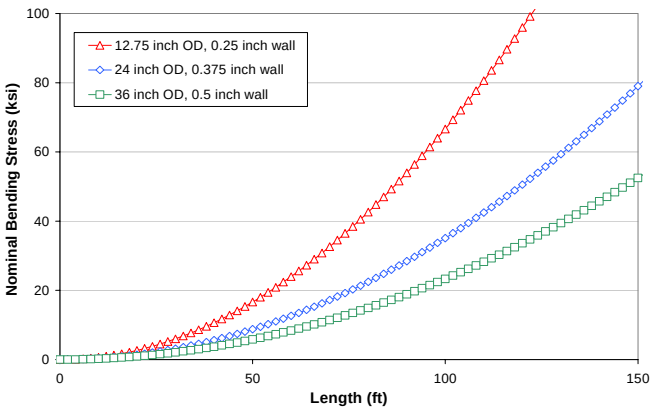

Figure 6 Plot of Case 3, cantilever sagging under own weight, calculated nominal bending stresses.

Unlike Cases 1 and 2, the figure demonstrates that the maximum bending stress is sensitive to pipe geometry. The smaller diameter pipe experiences larger stresses for a given length, as the smaller cross sectional and second moments of area cannot support the weight as effectively. The larger diameter pipe is heavier for a given length, but the larger cross sectional and second moments of area more effectively distribute the stress.

CASE 4. Fixed span.

This case is applicable when two (or more) sidebooms are used and the length of the pipeline between them sags under its own weight. Bending stresses are experienced at both of the sidebooms' lift points and at the center of the span. This case assumes that there is a significant length of pipeline outside of the sideboom lift points, such that the situation can be modeled effectively as a fixed beam as opposed to a freely supported span, which is given as Case 5. A schematic is given as Figure 7.

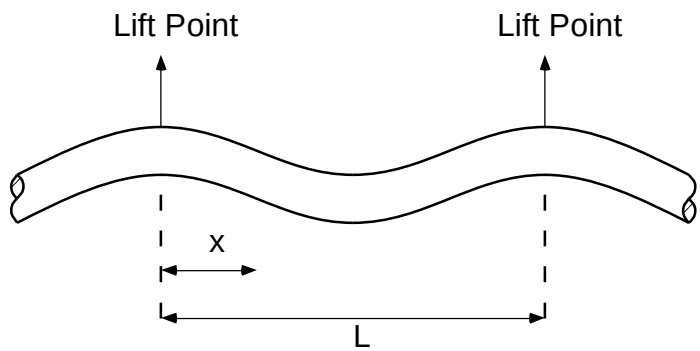

Figure 7 Schematic of Case 4. Fixed span.

As with the previous case the bending stresses in the beam are calculated as a function of span length, rather than height. It is assumed that the center of the span is not touching the ground. The maximum bending stress in the beam is given by:

$$\sigma_{\max} = \frac{1}{16} w L^2 \frac{y}{I} \qquad (10)$$

The stresses for this case are very sensitive to pipe geometry, as shown in Figure 8. The smaller diameter pipe experiences higher stresses than the larger diameter pipe. The lower second moment of area is not compensated for by the reduced cross-sectional area and associated weight.

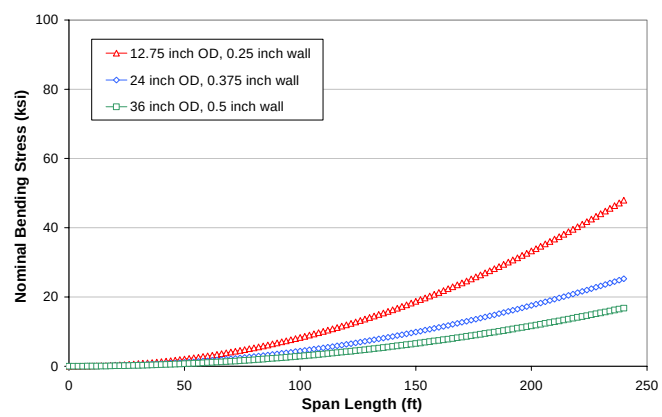

Figure 8 Plot of Case 4, fixed span, calculated nominal bending stresses.

CASE 5. Freely supported span.

This case is applicable when two (or more) sidebooms are used and the length of the pipeline between them sags under its own weight. This case assumes that the sideboom lift points are near the ends of the pipe and the situation can be modeled as a freely supported span. A schematic is given as Figure 9.

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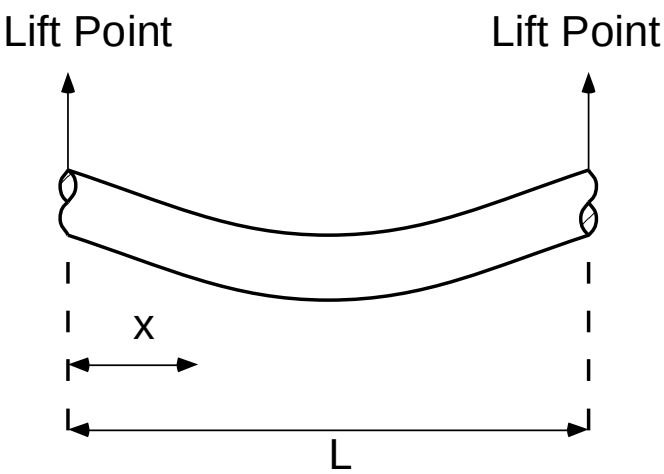

Figure 9 Schematic of Case 5. Freely supported span.

In this case, it is more applicable that the bending stresses in the beam are calculated as a function of span length, rather than height. It is assumed that the center of the span is not touching the ground. The maximum bending stress in the beam is given by:

$$\sigma_{\max} = \frac{1}{8} w L^2 \frac{y}{I} \qquad (11)$$

The stresses for this case are also very sensitive to pipe geometry, as shown in Figure 10.

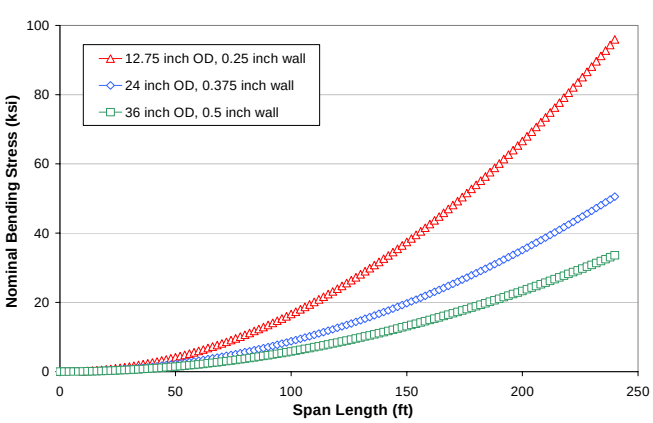

Figure 10 Plot of Case 5, freely supported span, calculated nominal bending stresses.

Note that the latter three cases are simple beam analyses that are selected from the literature [8]. It is only the former two cases in which any manipulation of the standard solutions is required. However, it is important to have the calculations for all possible configurations available for comparison purposes.

CASE 6. Torsion of the pipe.

It is possible that some torsion of the pipe is experienced during pipe laying operations, particularly if the pipe is moved laterally from the side of the trench and to the bottom of the trench. The torsional stresses are a function of the pipe geometry, the length of pipe and the rotation angle.

Consider a situation in which half of a pipeline is lying in the trench, and half is lying at surface level beside the trench (or on supports). The bending associated with this situation is comparable to Case 2, with the height of interest being the difference in height between the bottom of the trench and surface level (plus supports if present). The torsional stresses are a result of the pipe twisting sideways, see the schematic given as Figure 11.

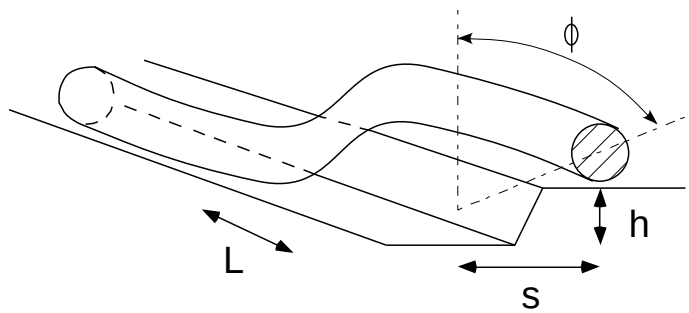

Figure 11 Schematic of Case 6. Torsion of the pipe.

The maximum shear stress in the pipe by the rotation is given by [8]:

$$\tau_{\max} = \frac{\phi \cdot G}{L} \frac{OD}{2} \qquad (12)$$

Where τ_{max} is the maximum shear stress, ϕ the rotation angle and G the shear modulus. The rotation angle is a function of both lateral distance from the side of the trench to the bottom center of the trench "*s*" and the height "*h*".

The polar moment of area "*J*" does not factor into this calculation, as the shear has been provided in terms of rotation angle. The shear is actually a function of applied torque "*T*", diameter and polar moment of area. The rotation angle is a function of torque, length polar moment and shear modulus. In combining the two relationships the polar moment falls out of the calculation. Therefore, there is no second moment "*I*" equivalent in the calculation. However, the torque necessary to apply a given rotation will be a function of the polar moment of area

The maximum shear is a function of both rotation angle and pipe length. This requires that there be two "critical" values considered in minimizing the shear. The relationship is plotted as Figure 12 for three common pipe geometries. For simplicity, it is assumed that the rotation angle is 45°, i.e. the lateral distance and height are comparable. If this is not the case for a

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given operation, the values must be adjusted accordingly using a ratio of actual angle divided by 45°.

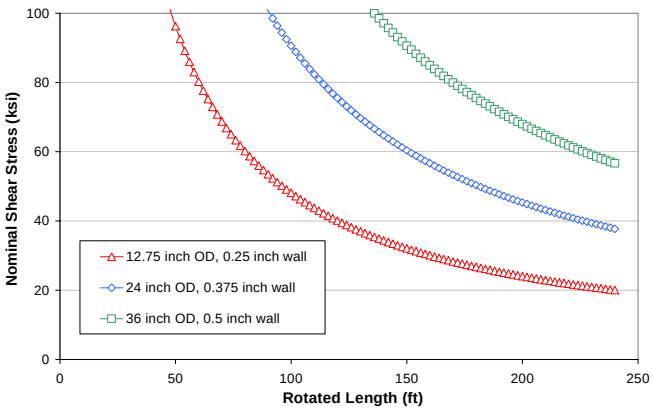

Figure 12 Plot of Case 6, torsion of the pipe, calculated nominal shear stresses.

Note that for this geometry long length of pipe is beneficial in reducing the shear stresses, as the rotation is distributed over a longer distance. In the other cases, in which bending stresses were calculated, length of pipe is detrimental to the calculated stresses as more distributed weight is involved.

The nominal shear stresses resulting from torsion of the pipe are sensitive to pipe geometry, as with Cases 3 through 5. Higher stresses are experienced by the larger diameter pipe, as the shear is a linear function of diameter.

It is important to recognize that in the case of pure shear, yielding will occur at approximately half the nominal yield strength due to the Tresca or Von Mises relationships. For example, a sample of steel with nominal yield strength of 52 ksi will yield at an applied nominal shear of 26 ksi. This must be considered in any assessments.

The analysis above does not account for the pipe twisting at both ends, as shown in Figure 11. If both ends twist, the shear stresses experienced will partially offset each other. The analysis provided above can therefore be considered as a conservative estimate.

EXAMPLE

In order to demonstrate the applicability of this work, a simple example is offered. Consider the pipe laying operation shown schematically as Figure 13. In this example a length of pipe is supported by two sidebooms. The pipe is 24 inch diameter with a 0.375 inch wall thickness. One end and its indefinite excess length of pipe is laying in the ditch. The central span is 200 feet between sideboom lift points. The other end is a 50 ft free length of pipe hanging beyond the second of the sidebooms. The lift points are 10 feet above the floor of the trench. It is assumed that the central span and free end of pipe are not touching the ground. Inspection of the schematic reveals that the various lengths of the pipe are comparable to Cases 2, 4 and 3.

The "S" curve to the left of the schematic in Figure 13 is comparable to Case 2. Inspection of Figure 4 indicates the nominal bending stress in the pipe is 56 ksi. The span in the center of the schematic is comparable to Case 4. Figure 8 gives the maximum nominal bending stress in a 200 ft span as 17.6 ksi. The free length of the pipe at the right is comparable to Case 3. Figure 6 gives the maximum nominal bending stress in a 50 ft cantilever as 8.8 ksi.

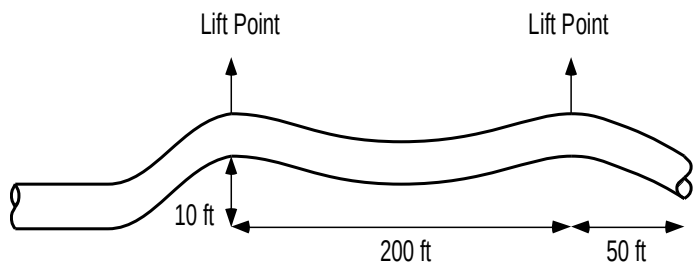

Figure 13 A schematic of a pipeline during a simple laying operation (not to scale).

The maximum estimated nominal bending stress in the pipeline is the 56 ksi experienced at the laid left end of the pipe. It appears that this is the critical stress for this particular geometry. If the steel were an X42 steel it would be expected to yield if lifted to this configuration. If the material were an X80, then it would not be expected to yield. Note that appropriate safety factors should be considered in any estimates.

The above work describes a simple technique for estimating the bending stress in pipeline during laying operations. It is based on some simple assumptions that have been discussed during the derivation of the various equations. The six cases discussed are idealized and not necessarily representative of the load and deflection of any specific operation. It is expected that the lengths of pipeline involved in an in-service operation can be broadly categorized to allow guidance. These six cases do not consider varying lift heights at the lift points, and this is a possible direction for future work. The equations used have not been verified by field trials so should not be directly relied upon. Future work may demonstrate their correctness, or they may be modified to account for unforeseen circumstances that were not identified in this analysis.

SUMMARY / CONCLUSION

Estimates of the nominal bending and torsional stresses in pipe that are experienced during laying operations have been derived or presented. These stresses are calculated as a function of pipe geometry and lift height or span length. The calculated values can be compared against the yield strength of pipe material to estimate the susceptibility of the pipe to failure

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during laying operations. Safety factors are not provided so should be considered as appropriate.

The work provides simple guidelines that can be used to improve laying protocols and reduce damage during pipeline construction. It may lead to the development of more formalized standards.

NOMENCLATURE

A = cross-sectional area of pipe
E = Young's modulus
G = shear modulus
I = second moment of inertia
J = polar moment of inertia
ID = inner diameter of pipe
L = length of pipe
M = moment
OD = outer diameter
s = distance pipeline moved sideways
t = wall thickness
T = torque
w = weight/unit length
x = distance along beam
y = distance to neutral axis
δ = deflection
ν = Poisson's ratio
ρ = density
σ = normal stress
τ = shear stress

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